

Algebra II with Trigonometry

Chapter 6.1

Olympiad Problems (GPT-5.4)

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Problems

1. Let P be the terminal point on the unit circle determined by t , where t is chosen so that

$$\cos t + \sin t = \frac{1}{3} \quad \text{and} \quad P \text{ lies in Quadrant II.}$$

Find the exact coordinates of P .

2. A point $P(x, y)$ lies on the unit circle in Quadrant I and satisfies

$$x - y = \frac{1}{5}.$$

Find $x + y$.

3. Suppose the terminal point determined by t is $P = (x, y)$ on the unit circle. If the terminal point determined by $2t$ is $\left(-\frac{7}{9}, \frac{4\sqrt{2}}{9}\right)$, and P lies in Quadrant II, find P .

4. Find the exact value of

$$\cos \frac{\pi}{12} + \sin \frac{\pi}{12} + \cos \frac{5\pi}{12} + \sin \frac{5\pi}{12}.$$

5. Let P be the terminal point determined by t on the unit circle. If the three points determined by t , $\pi - t$, and $-t$ have x -coordinates whose sum is 1, find all possible y -coordinates of P .

6. A point $P = (x, y)$ on the unit circle has the property that reflecting P across the line $y = x$ gives the terminal point determined by $t + \frac{\pi}{6}$, where P itself is determined by t . Find all possible exact coordinates of P .